Zabbix SNMP Monitoring Lab on PNETLab

Lab report, configuration reference and troubleshooting log

Repository: zabbix-pnetlab-monitoring

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1. Summary

This project builds a small virtual network in PNETLab and monitors it from a Zabbix server using SNMPv2c. The Zabbix server is a VMware appliance outside the lab. It reaches the whole topology through a single link to the PNETLab management cloud (Cloud0), which is attached to one router (R5). Every other device is polled through that router.

The report documents the topology and addressing, the exact configuration of every device, how monitoring was set up in Zabbix, the failure tests used to prove it works, and, most importantly, every problem encountered along the way with its diagnosis and fix.

Result. Four devices (R5, R6, SW1, SW2) are monitored over SNMP with a green availability status, interface and device metrics are collected, and two controlled failures (an interface shutdown and a link cut) are detected and reported by Zabbix.

2. Environment

Component	Details
Network emulator	PNETLab, VM at 192.168.139.128
Hypervisor	VMware Workstation on a Linux host; NAT network vmnet8 (192.168.139.0/24, gateway 192.168.139.2)
Monitoring server	Zabbix 7.4.15 VMware appliance, 192.168.139.129
Lab devices	Cisco IOL routers (R5, R6) and IOL L2 switches (SW1, SW2), two VPCS end hosts
Protocol	SNMPv2c, read-only community, Zabbix template 'Cisco IOS by SNMP'

3. Topology and addressing

Cloud0 is connected only to R5 (e0/2). The appliance sits on the same 192.168.139.0/24 network as Cloud0, so it reaches R5 directly and every other device through R5. This keeps the lab self-contained and mirrors how a real monitoring server reaches remote devices through routing.

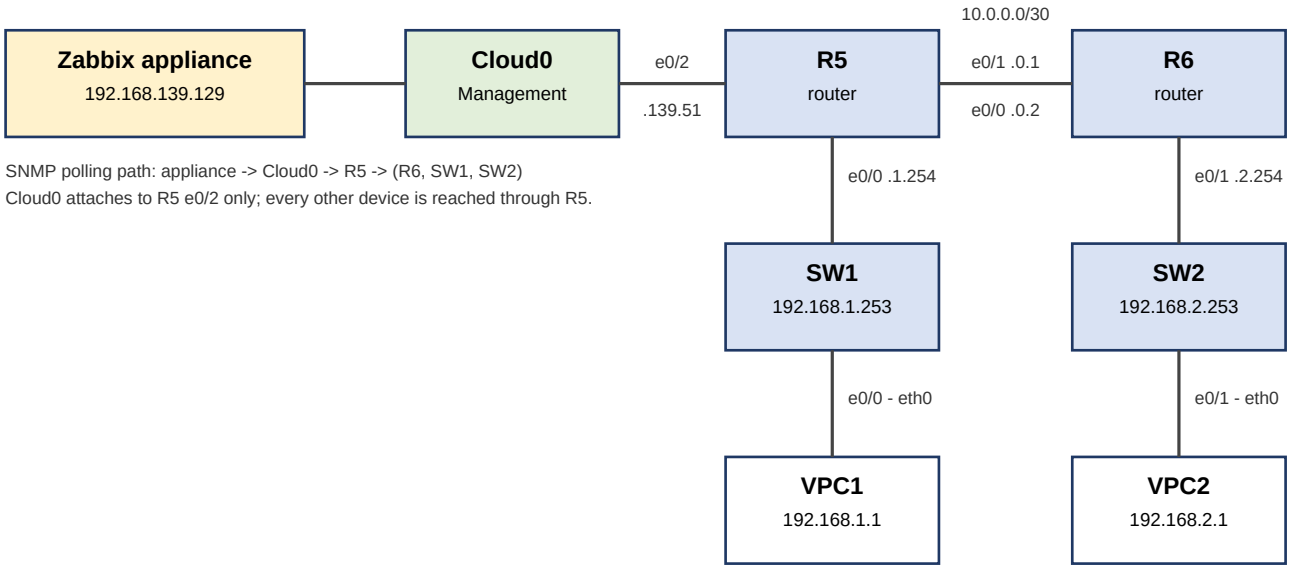


Figure 1. Logical topology (node labels in PNETLab: R5, R6, SW1, SW2, VPC1, VPC2).

Device	Interface	Address	Role
R5	e0/0	192.168.1.254/24	Gateway of LAN 1, link to SW1 e0/1
R5	e0/1	10.0.0.1/30	Link to R6
R5	e0/2	192.168.139.51/24	Management link to Cloud0 (polled by Zabbix)
R6	e0/0	10.0.0.2/30	Link to R5 (polled by Zabbix)
R6	e0/1	192.168.2.254/24	Gateway of LAN 2, link to SW2 e0/0
SW1	Vlan1	192.168.1.253/24	Management IP, gateway 192.168.1.254
SW2	Vlan1	192.168.2.253/24	Management IP, gateway 192.168.2.254
VPC1	eth0	192.168.1.1/24	End host, gateway 192.168.1.254
VPC2	eth0	192.168.2.1/24	End host, gateway 192.168.2.254
Appliance	eth0	192.168.139.129/24	Zabbix server, static routes via R5

4. Device configuration

The configuration below is the final working state.

4.1 R5 (the only router attached to Cloud0)

```
hostname R5
interface e0/0
 ip address 192.168.1.254 255.255.255.0
 no shutdown
interface e0/1
 ip address 10.0.0.1 255.255.255.252
 no shutdown
interface e0/2
 ip address 192.168.139.51 255.255.255.0
 no shutdown
ip route 192.168.2.0 255.255.255.0 10.0.0.2
snmp-server community public RO
```

4.2 R6

```
hostname R6
interface e0/0
 ip address 10.0.0.2 255.255.255.252
 no shutdown
interface e0/1
 ip address 192.168.2.254 255.255.255.0
 no shutdown
ip route 192.168.1.0 255.255.255.0 10.0.0.1
ip route 192.168.139.0 255.255.255.0 10.0.0.1
snmp-server community public RO
```

The route to 192.168.139.0/24 is what lets R6 and SW2 answer the Zabbix server.

4.3 SW1 and SW2 (shown for SW2; SW1 uses 192.168.1.253 and gateway 192.168.1.254)

```
hostname SW2
no ip routing
interface vlan 1
 ip address 192.168.2.253 255.255.255.0
 no shutdown
ip default-gateway 192.168.2.254
snmp-server community public RO
```

Important: 'no ip routing' is required on these IOL switches. With routing enabled the switch ignores 'ip default-gateway' (see Problem 6). The alternative is to keep routing on and add 'ip route 0.0.0.0 0.0.0.0 192.168.2.254'.

4.4 End hosts (VPCS)

```
ip 192.168.1.1/24 192.168.1.254     (VPC1)
ip 192.168.2.1/24 192.168.2.254     (VPC2)
```

4.5 Zabbix appliance routes

The appliance's default gateway is the VMware NAT gateway, which knows nothing about the lab. Static routes through R5 are needed:

```
ip route add 192.168.1.0/24 via 192.168.139.51
ip route add 192.168.2.0/24 via 192.168.139.51
ip route add 10.0.0.0/30    via 192.168.139.51

# make them permanent (NetworkManager)
nmcli con modify <connection> +ipv4.routes "192.168.1.0/24 192.168.139.51"
nmcli con modify <connection> +ipv4.routes "192.168.2.0/24 192.168.139.51"
nmcli con modify <connection> +ipv4.routes "10.0.0.0/30 192.168.139.51"
nmcli con up <connection>
```

4.6 Host (VMware) permission for the lab network

PNETLab routers use virtual MAC addresses (aa:bb:cc:...). VMware only delivers frames for those MACs to the PNETLab VM when the VM can use promiscuous mode on the virtual network. On a Linux host this is controlled by the permission on the vmnet device:

```
ls -l /dev/vmnet8               # expect crw-rw-rw-
sudo chmod a+rw /dev/vmnet8     # then power on the VMs
```

The permission is read when a VM powers on, and it resets when the host reboots. A systemd oneshot service that runs the chmod at boot (before the VMs start) makes it persistent.

5. Zabbix configuration

Hosts were created under Data collection > Hosts > Create host, using the template **Cisco IOS by SNMP**, an SNMP interface on port 161, SNMP version 2 and the community macro {$SNMP_COMMUNITY} (default 'public'). New host groups are created by typing their name in the host group field.

Host	Group	SNMP interface IP	Path from the appliance
R5	Routers	192.168.139.51	Direct (Cloud0)
R6	Routers	10.0.0.2	Via R5
SW1	Switches	192.168.1.253	Via R5
SW2	Switches	192.168.2.253	Via R5, then R6

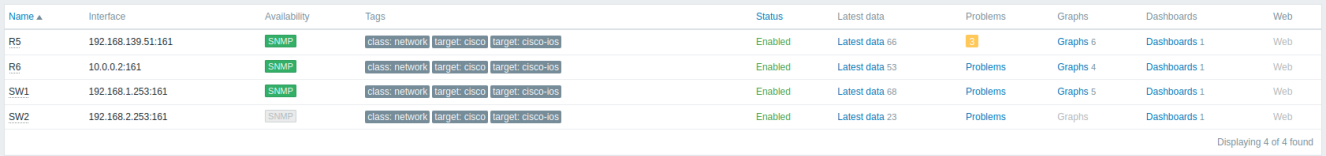


Figure 2. All four devices with a green SNMP availability badge. (SW2 was still grey, waiting for its first poll, when this screenshot was taken.)

6. Validation and failure tests

Connectivity checks from the appliance before adding hosts: ping 192.168.139.51, 10.0.0.2, 192.168.1.253 and 192.168.2.253 all returned 0% packet loss.

Test 1 - interface shutdown. On R6, interface e0/1 (toward SW2) was shut down.



Name ▲	Interface	Availability	Tags	Status	Latest data	Problems	Graphs	Dashboards	Web
R5	192.168.139.51:161	SNMP	class: network target: cisco target: cisco-ios	Enabled	Latest data 66	3	Graphs 6	Dashboards 1	Web
R6	10.0.0.2:161	SNMP	class: network target: cisco target: cisco-ios	Enabled	Latest data 46	1	Graphs 5	Dashboards 1	Web
SW1	192.168.1.253:161	SNMP	class: network target: cisco target: cisco-ios	Enabled	Latest data 68	Problems	Graphs 5	Dashboards 1	Web
SW2	192.168.2.253:161	SNMP	class: network target: cisco target: cisco-ios	Enabled	Latest data 68	Problems	Graphs 5	Dashboards 1	Web

Displaying 4 of 4 found

Figure 3. Test 1: SW2 is unreachable (red); R6 stays green and reports a problem (orange); R5 and SW1 are unaffected.

This is the expected result. Zabbix polls R6 at 10.0.0.2 over the R5-R6 link, which was not touched, so R6 stays reachable and reports its own interface problem. Only SW2, which sits behind the failed interface, becomes unreachable. The device nearest the fault raises the alarm; the device behind it disappears.

Test 2 - link cut. On R5, interface e0/1 (toward R6) was shut down, cutting the path to both R6 and SW2.

Device	Expected result in Zabbix
R5	Green; 'Link down' problem for the failed interface
R6	Unreachable (red) after several missed polls
SW2	Unreachable (red) after several missed polls
SW1	Green, unaffected

After 'no shutdown' the devices returned to green and the problems resolved. Screenshots of this test are stored in the repository's screenshots folder.

7. Problems encountered and how they were solved

The problems are listed in the order they were found. Several look unrelated at first but are all consequences of one idea: a device only answers what it has a route (or a path) back to. Most of the time went into isolating *which hop* was failing.

Problem 1: PCs could not ping each other across the routers

Symptom. Pings from a VPC returned 'ICMP type:3 ... unreachable' from its own gateway.

Cause. Each router only knew its directly connected networks. R5 had no route to 192.168.2.0/24 and R6 had none to 192.168.1.0/24.

How it was diagnosed. The unreachable message came from the gateway address (192.168.1.254 / .2.254), meaning the first router received the packet and had no route for it. 'show ip interface brief' showed all interfaces up/up, so the issue was routing, not links.

Fix. Add static routes on both routers (a route is needed in both directions, because the reply must also find its way back). OSPF would work equally well.

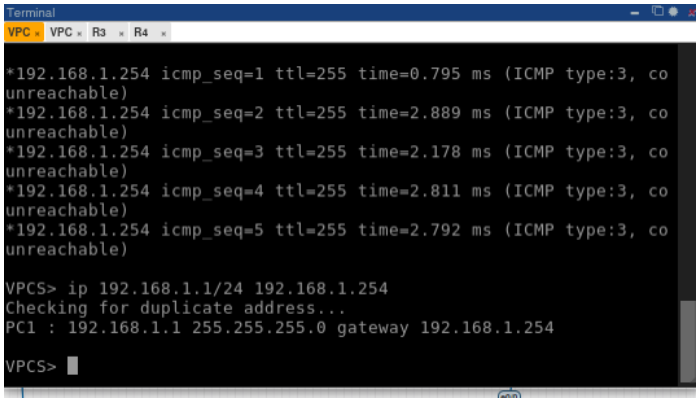


Figure 4. Symptom: the gateway answers with ICMP destination unreachable.

Problem 2: Zabbix appliance could not ping the routers, but PNETLab could

Symptom. Appliance to 192.168.139.51/.52: 100% loss. PNETLab to the same IPs: 0% loss. Router to PNETLab (.128): 80% success. Router to the VMware gateway (.2): failed.

Cause. VMware did not deliver unicast frames addressed to the routers' virtual MACs to the PNETLab VM, because the vmnet8 device had no permission for promiscuous mode (/dev/vmnet8 was crw-------). Broadcast traffic such as ARP still worked, which made it look like a layer 2 success.

How it was diagnosed. Layer by layer. The router's ARP table did contain the appliance's MAC, and the appliance's neighbour table contained the router MACs (so ARP crossed). Then a capture on the host's vmnet8 interface showed echo requests from the appliance but no replies, while a capture inside PNETLab (pnet0) showed nothing at all. That proved the frames were being dropped between the virtual switch and the PNETLab VM.

Fix. Run 'sudo chmod a+rw /dev/vmnet8' on the host, then fully power off and power on PNETLab (first) and the appliance. VMware reads the permission only when a VM powers on, so restarting the guest operating system is not enough. The fix resets when the host reboots; a boot-time service makes it permanent.

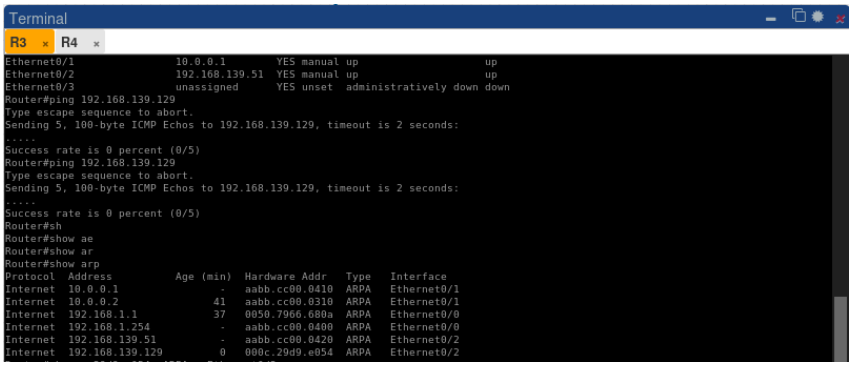


Figure 5. The router resolved the appliance's MAC (000c.29d9.e054) but pings still failed: a one-way, unicast-only drop.

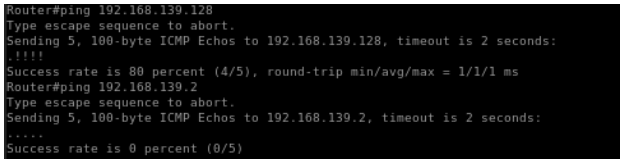


Figure 6. Router to PNETLab works; the gateway at .2 does not reply to pings (normal for the VMware NAT gateway).

Problem 3: The same fix seemed to be needed again in a new topology

Symptom. A fix applied in an earlier lab no longer worked.

Cause. Promiscuous-mode permission is a host/VMware setting, not a per-topology setting. It resets after a host reboot, so a previously fixed lab looks broken again.

How it was diagnosed. Comparing: the permission on /dev/vmnet8 had returned to its default.

Fix. Check 'ls -l /dev/vmnet8' before any Zabbix lab, and automate the chmod at boot.

Problem 4: Packet captures run on the wrong machine

Symptom. 'tcpdump -ni pnet0' on the host returned 'No such device exists'.

Cause. pnet0 exists only inside the PNETLab VM. The host's equivalent interface is vmnet8.

How it was diagnosed. Error message. Lesson: know which machine owns each interface before capturing.

Fix. Capture on vmnet8 on the host and on pnet0 inside PNETLab, then compare what each side sees.

Problem 5: SNMP community typo and missing tools

Symptom. A command typed as 'snmp-server community public R0' (zero instead of the letter O) and 'snmpwalk: command not found' on the appliance.

Cause. RO (read-only) is two letters, not a zero. The appliance does not ship net-snmp-utils.

How it was diagnosed. 'show run | include snmp' on the device reveals the exact line. The missing package is reported by the shell.

Fix. Re-enter the line as 'snmp-server community public RO'. Install the tools if manual testing is needed: 'dnf install net-snmp-utils'.

Problem 6: SW2 answered from its own subnet but not from other networks

Symptom. R6 could ping SW2 (192.168.2.253) but R5 could not, and SW2 could not ping 10.0.0.1, even though routes and the default gateway looked correct.

Cause. IP routing was enabled on the IOL switch. In that mode the switch uses a routing table and ignores 'ip default-gateway', so it had no route back to 10.0.0.1. R6 worked only because it shares SW2's subnet: no gateway is needed to reply to a directly connected network.

How it was diagnosed. Hop-by-hop pings isolated the problem to the last hop's return path. 'show ip route' on SW2 printed a table with 'Gateway of last resort is not set' and only the connected 192.168.2.0/24 route. An empty 'show run | include ip routing' proved nothing, because this image enables routing by default and does not print the line.

Fix. Turn routing off ('no ip routing') so the existing default gateway takes effect, or keep routing on and add 'ip route 0.0.0.0 0.0.0.0 192.168.2.254'.

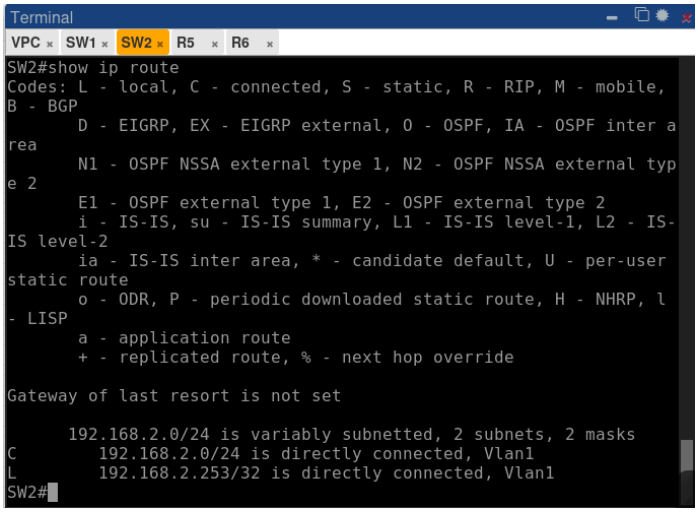


Figure 7. SW2 had routing enabled and no default route: it only knew its own subnet.

Problem 7: Appliance had no route into the lab

Symptom. From the appliance: R5 answered, but 10.0.0.2 and 192.168.2.253 timed out, and 192.168.1.253 returned 'Destination Host Unreachable' from an unrelated 192.168.1.x address.

Cause. The appliance's default gateway is the VMware NAT gateway, which does not know the lab subnets. Packets for 192.168.1.0/24 were even forwarded toward the real home network, which uses the same range.

How it was diagnosed. 'ip route get <address>' showed every lab destination leaving via 192.168.139.2 instead of via R5 (192.168.139.51).

Fix. Add static routes for the lab subnets via R5 and make them persistent with NetworkManager (see section 4.5). Caveat: overlapping the lab range with a real LAN makes that real LAN unreachable from the appliance; using a non-overlapping range such as 172.16.x.x avoids this.

Problem 8: Zabbix web interface: 'Database error - Connection refused'

Symptom. The Zabbix login page showed a database error after the appliance was restarted.

Cause. The web front end was running but the database service was not accepting connections after the VM restart (probable cause: services not running after the power cycle).

How it was diagnosed. Check the database service status, free disk space and memory, and the service journal from the appliance console or SSH.

Fix. Start the database service, restart the Zabbix server and web services, then reload the page. Shut the appliance down cleanly with 'poweroff' to avoid database damage.

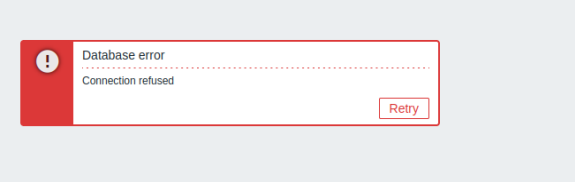


Figure 8. Database error shown by the Zabbix front end.

Problem 9: Old hosts left over from a previous lab

Symptom. Hosts named router1 (192.168.139.50) and SW1 (10.10.1.2) were red in Zabbix.

Cause. They belonged to an earlier topology that no longer existed.

How it was diagnosed. The hosts list showed addresses that were not in the current topology.

Fix. Delete the obsolete hosts (Data collection > Hosts) and create the new ones. Do not delete the built-in 'Zabbix server' host or the default host groups.

Problem 10: Warnings 'Interface Et0/x: In half-duplex mode' on R5

Symptom. Three warning-level problems appeared on R5 even though it was healthy.

Cause. The Cisco IOS template raises a warning when an interface reports half duplex. The emulated IOL Ethernet interfaces report half duplex, so this is a false positive in a lab.

How it was diagnosed. The Problems page lists the exact trigger names (one per interface), and the interfaces were up and passing traffic.

Fix. Safe to ignore in a lab, or reduce the noise by disabling that trigger prototype or tuning the macro for the template. On real hardware this warning would indicate a duplex mismatch worth investigating.

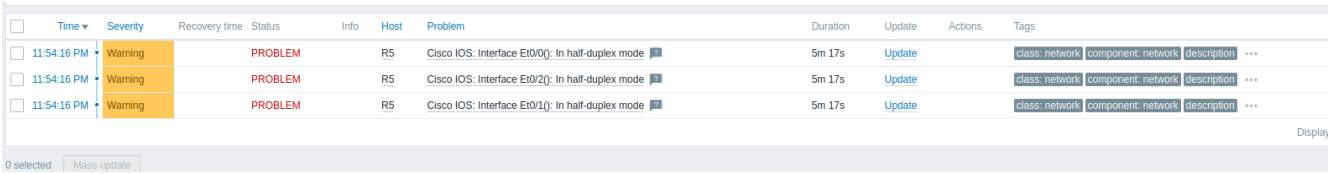


Figure 9. Half-duplex warnings reported for R5's interfaces.

8. Troubleshooting method used

The fastest way through every problem above was to test one hop at a time and to ask whether the failure was on the way out or on the way back.

Layer / question	Test	Typical conclusion
Is the interface up and configured?	show ip interface brief	up/up with the right IP; up/up alone does not prove a cable
Is there layer 2 reachability?	show arp / ip neigh	MAC listed = ARP crossed; says nothing about unicast delivery

Layer / question	Test	Typical conclusion
Does traffic leave the source?	tcpdump on the source side	Requests seen = source is fine
Does traffic reach the next segment?	tcpdump on the next interface	Missing = dropped in between (VMware, cable, filter)
Is there a route there and back?	show ip route / ip route get	Missing return route is the most common cause of one-way failures
Does the device answer from other subnets?	Ping from different sources	Works locally but not remotely = return path / default gateway
Does SNMP itself work?	show run | include snmp; snmpwalk; Zabbix SNMP badge	Ping OK but badge red = community, version or ACL

9. Lessons learned

1. A successful ARP exchange does not prove unicast traffic works; broadcast and unicast can be treated differently by a virtual switch.
2. When a device answers from its own subnet but not from others, suspect its return path first.
3. A switch with IP routing enabled ignores 'ip default-gateway'. Check 'show ip route' to see which mode a device is in.
4. Host-side settings (VMware permissions, appliance routes) are part of the network and must be documented and automated.
5. A monitoring server reaching devices in other subnets needs routes of its own, not just routes on the devices.
6. Monitoring shows where the failure is: the device nearest the fault alarms, the devices behind it become unreachable.

10. Possible improvements

Use SNMPv3 (authentication and encryption) instead of a plain-text community string; restrict SNMP with an access list; monitor the VPCs with the ICMP Ping template; add maps and a dashboard; configure email or messaging alerts; renumber the lab to a range that does not overlap a home network; add a DHCP server and a redundant core to extend the topology.